

Shanghai United International School (Hongqiao) · 5-Minute Daily Math Warmup

Classic Error Practice & Teacher Mnemonics Flashcards

Student Name: _____

Cycle: Semester 1, Weeks 3-4 Daily Goal: Complete 2 cards per day, master the core concepts

Card 01

Divisibility Modulo · William's Focus

💡 **Mnemonic: Split 72 into 8 and 9! Last 3 digits determine Y, sum of all digits determines X. Complete both steps!**

[Parallel Practice Challenge]

The six-digit number $\overline{73A64B}$ is divisible by both 8 and 9. Find A and B .

① Last 3 digits: $\overline{64B} \div 8$ has no remainder, so $B =$ _____;

② Sum of digits: $7 + 3 + A + 6 + 4 + B$ is a multiple of 9, so $A =$ _____.

[Key]: $B = 0$ or 8 . If $B = 0$, $A = 7$ (737640); if $B = 8$, $A = 8$ (738648).

Card 02

Decimal Addition & Division · William

💡 **Mnemonic: Ten tenths make one whole! $0.9 + 0.1 = 1.0$. In division, shift the decimal point equally: $5.4 \div 0.6 = 54 \div 6 = 9$!**

[Mental Speed Drill]

(1) $0.8 + 0.2 =$ _____

(2) $4.9 + 0.1 =$ _____

(3) $7.2 \div 0.8 =$ _____ (⚠ NOT 0.9!)

(4) $40 \div 0.5 =$ _____

[Key]: (1) 1.0; (2) 5.0; (3) 9; (4) 80.

Card 03

Powers vs Products · Both Students

💡 **Mnemonic: Base multiplies itself! 2^3 means three 2s multiplied: $2 \times 2 \times 2 = 8$, NEVER $2 \times 3 = 6$!**

[Write Out Full Expanded Form]

(1) $2^3 =$ _____ $=$ _____

(2) $3^4 =$ _____ $=$ _____

(3) $10^3 \times 10^4 = 10^{+} = 10^{+}$

[Key]: (1) $2 \times 2 \times 2 = 8$; (2) $3 \times 3 \times 3 \times 3 = 81$; (3) 10^7 .

Card 04

Square Numbers · Sophia's Focus

💡 **Mnemonic: Twin identical factors make a square! $2 \times 3 = 6$ is a rectangle, so 6 is NEVER a square!**

[Identification & Geometry]

(1) Circle the square numbers:

12, 16, 18, 25, 27, 36, 45, 49

(2) A square has an area of 49 cm^2 . What is its side length?

Answer: Side length = _____ cm (Do NOT write $49 \div 2$).

[Key]: (1) Square numbers: 16, 25, 36, 49; (2) Side length = 7 cm ($7 \times 7 = 49$).

Card 05

Fraction Expansion · Sophia's Focus

💡 Mnemonic: What you do to the bottom, you **MUST** do to the top! Never leave the numerator unscaled!

[Find and Fix the Error]

Mistake: $\frac{2}{3} + \frac{1}{4} = \frac{2+1}{12} = \frac{3}{12} = \frac{1}{4}$

Correct solution:

$$\frac{2}{3} + \frac{1}{4} = \frac{\quad}{12} + \frac{\quad}{12} = \underline{\hspace{2cm}}$$

[Key]: Expression = $\frac{8}{12} + \frac{3}{12} = \frac{11}{12}$.

Card 06

Smart Fraction Comparison

💡 Mnemonic: Equal numerators? Smaller denominator is bigger! Close to 1? Smaller residual distance is bigger!

[Fill with > or <, state rule]

(1) $\frac{9}{16}$ _____ $\frac{9}{20}$ (Same numerators)

(2) $\frac{19}{20}$ _____ $\frac{29}{30}$ (Compare distance to 1)

[Key]: (1) >; (2) < ($1 - \frac{19}{20} = \frac{1}{20} > 1 - \frac{29}{30} = \frac{1}{30}$).